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Review Article

Efficacy of Lactobacillus Reuteri in Improving Gut Barrier Function & Management of Infantile Colic

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Abstract: Infantile colic is a frustrating impasse that affects up to 20% of infants. Nevertheless, its pathogenesis is currently unknown, some hypotheses are food hypersensitivity or allergy, gut dysmotility, inflammation, and visceral pain. Probiotics use in treatment and prevention of infantile colic is a moderately new topic. Moreover, Probiotics strains exhibit a wide range of health benefits by modifying the intestinal microbiota and immunity. Lactobacillus reuteri is one of the utmost widely studied probiotic strains. It endorses gut health by stimulation of mucosal gut barrier functions, production of antimicrobial substances (such as reuterin and lactic acid) and influencing acquired and innate immune responses. Reuterin produced by L. reuteri is a potent anti-microbial compound capable of inhibiting a wide spectrum of pathogenic microorganisms. L. reuteri creates biofilms that stimulate tumor necrosis factor production by lipopolysaccharide (LPS)-activated monocytoid cells, apart from antimicrobial metabolite production. Interestingly, as a potential therapy for childhood, L. reuteri administration has emerged functional gastrointestinal disturbances as these disturbances are associated with gut microbiota perturbations in early life. The review précises the beneficial aspects of the probiotic L. reuteri strain in clinical practice with a special focus on its role in improving gut health and immunity including managing Infantile colic in infants and toddlers.

Keywords: Lactobacillus Reuteri, Infantile Colic, Functional Gastrointestinal